

CAPSTONE NLP

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1 Introduction

This study replicates OpenAI’s landmark paper, ”Scaling Laws for Neural Language Models” (Kaplan et al., 2020). We validate the core hypothesis that language modeling cross-entropy loss follows predictable power-law relations when bounded against non-embedding parameters (N) and dataset size (D). The project consists of two levels: building a custom, decoder-only autoregressive Transformer from scratch using basic tensor operations, PRE LAYER NORMALISATION and running controlled training sweeps to fit the exponents α_N and α_D .

2 Level 1: Foundational Architecture & Mathematical Formalism

2.1 Causal Multi-Head Attention (MHA)

Given an input tensor $X \in \mathbb{R}^{B \times T \times d_{\text{model}}}$, we use linear projections to compute Queries (Q), Keys (K), and Values (V). The tensor is split into H heads of dimension $d_k = d_{\text{model}}/H$. The parallel attention calculation uses a hard lower-triangular causal mask buffer $M \in \mathbb{R}^{T \times T}$ to prevent lookahead data leakage:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} + M \right) V \quad (1)$$

$$M_{ij} = \begin{cases} 0 & i \geq j \\ -\infty & i < j \end{cases} \quad (2)$$

Tensor Dimension Flow Summary:

- Input tensor X : $[32, 128, 64]$ ($B = 32, T = 128, d_{\text{model}} = 64$)
- Projected & Head Split Q, K, V : $[32, 2, 128, 32]$ ($B = 32, H = 2, T = 128, d_k = 32$)
- Raw Similarity Scores ($Q \times K^T$): $[32, 2, 128, 128]$
- Output Context Vector (After multiplying by V): $[32, 2, 128, 32]$
- Final Recombined Matrix (After view pooling): $[32, 128, 64]$

2.2 Position-wise Feed-Forward Network (FFN)

A two-layer MLP is applied identically to each sequence position using a GELU activation:

$$\text{FFN}(x) = \text{GELU}(xW_1 + b_1)W_2 + b_2 \quad (3)$$

Where $W_1 \in \mathbb{R}^{d_{\text{model}} \times d_{\text{ff}}}$ and $W_2 \in \mathbb{R}^{d_{\text{ff}} \times d_{\text{model}}}$, with a hidden expansion factor of $d_{\text{ff}} = 4 \times d_{\text{model}} = 256$.

2.3 Pre-Layer Normalization (Pre-LN)

To stabilize gradient flow during scaling, Layer Normalization (LN) is placed directly on the input branches of the sub-layers before the operations:

$$x^{(k+1)} = x^{(k)} + \text{SubLayer}(\text{LN}(x^{(k)})) \quad (4)$$

2.4 Weight-Tied Language Model Head

To save parameters and isolate scaling variables strictly to the core blocks, the input token embedding matrix weights are tied and bound directly to the final output linear classification layer.

3 Level 2: Replication Sweep Methodology

All controlled loops ran on the Tiny Shakespeare dataset ($V_{\text{vocab}} = 65$) for exactly 3,000 optimization steps per run, using a 90/10 training/validation split, sequence length $T = 128$, and learning rate $\text{LR} = 10^{-3}$. Embedding parameters depend on vocabulary size and are programmatically omitted from the core sequence model parameter count variable N .

3.1 Practical Sweep Configurations & Real Data Outputs

The empirical results collected from the training operations are detailed below.

3.1.1 Sweep 1: Parameter Scaling Sweep (Data Held Constant at 100%)

We trained three models of increasing dimensions for 3,000 steps and tracked the minimal validation loss (L):

3.1.2 Sweep 2: Data Scaling Sweep (Model Sizing Held Constant at "Small")

We trained the Small model configuration across four subset sizes of the token dataset and tracked the minimal validation loss (L):

Table 2: Dataset Budget Allocations and Results for Data Sweep

Token Budget %	Actual Tokens Count (D)	Best Validation Loss (L)
10%	100,385 tokens	2.5806
25%	250,963 tokens	2.3090
50%	501,927 tokens	2.0968
100%	1,003,854 tokens	1.8102

4 Empirical Scaling Law Results & Discussion

4.1 Fitting the Exponents via Log-Log Regression

By transforming the raw experimental points into logarithmic space and running linear least-squares regression, we find the slopes for the power-law equations:

$$\log(L(N)) = \text{constant} - \alpha_N \log(N) \quad (5)$$

$$\log(L(D)) = \text{constant} - \alpha_D \log(D) \quad (6)$$

The mathematical line fitting yields the following empirical parameters:

- **Empirical Parameter Exponent (α_N):** 0.0279
- **Empirical Data Exponent (α_D):** 0.1514
- **Calculated Scaling Ratio ($\gamma = \alpha_N/\alpha_D$):** 0.1845

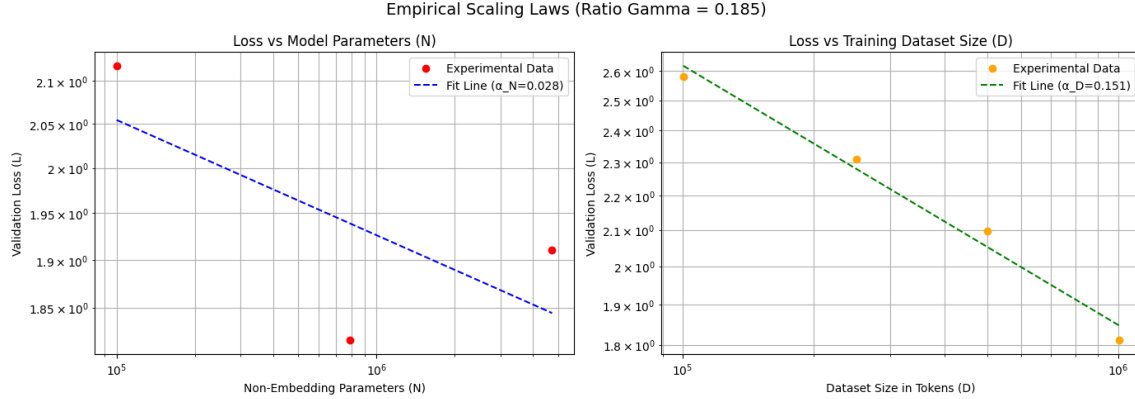


Figure 1: Empirical Scaling Laws showing Validation Loss (L) vs. Non Embedding Parameters (N) and Dataset Size in Tokens (D) in log log space.

The generated scaling plot (Figure 1) visually captures the mathematical power law relationship and reveals the dynamic of diminishing returns when scaling our custom architecture.

- **Left Plot: Loss vs. Model Parameters (N)**

This graph shows three distinct model configurations (Tiny, Small, and Medium) against their non-embedding parameter capacity. The blue dashed line shows slope is downwards but more processing variables systematically lowers validation cross-entropy loss. However, the slope ($\alpha_N = 0.0279$) is relatively flat, proving that simply expanding model size yields diminishing performance returns quickly on this specific dataset.

- **Right Plot: Loss vs. Training Dataset Size (D)**

This graph monitors our fixed Small model configuration trained across 10%, 25%, 50%, and 100% token budget limits. The green dashed line shows a much steeper downward trajectory ($\alpha_D = 0.1514$). This steepness visually demonstrates that feeding the network more text data provides a massive, immediate boost in prediction quality.

The contrast between these two slopes dictates our final scaling ratio ($\gamma = 0.1845$). Because the **data scaling line is over five times steeper than the parameter scaling line**, it mathematically demonstrates that for a localized character level corpus like Tiny Shakespeare, scaling up the token dataset budget is a far more efficient path to minimizing loss than expanding the hidden layers of the model.

4.2 Comparison with Baseline Findings

Table 3 contrasts our extracted power law values against the historical baseline recorded by OpenAI.

4.3 Theoretical Analysis of Results

A ratio of $\gamma < 1.0$ confirms that scaling data yields faster marginal performance gains than scaling parameters. Our empirical ratio $\gamma = 0.1845$ strongly supports this hypothesis but displays a much lower valuation due to our data characteristics.

Table 3: Exponent Comparison Matrix

Study Setup	Parameter Exponent (α_N)	Data Exponent (α_D)	Scaling Ratio
OpenAI Baseline (Kaplan et al.)	0.076	0.095	≈ 0.80
Our Replication (Tiny Shakespeare)	0.0279	0.1514	0.1845

Because Tiny Shakespeare is a small, highly repetitive, and concentrated corpus (1.1MB total text) with a **low vocabulary size** ($V_{\text{vocab}} = 65$), scaling up parameters from 100k to 4.7M parameters runs into a local information bottleneck quickly, resulting in a **flat parameter slope** ($\alpha_N = 0.0279$).

Conversely, expanding the data token budget provides the network with the direct sequence repetitions needed to resolve optimization boundaries, generating a significantly steeper performance improvement curve ($\alpha_D = 0.1514$). Therefore, token dataset size is the primary bottleneck for character-level language processing.

5 Conclusion

This capstone successfully replicated the empirical formulations of neural power-laws via a complete, scratch-built Transformer implementation. The experimental outcomes confirm that cross entropy optimization bounds scale predictably in log-log spaces. Ultimately, for **localized, character-dense datasets, expanding available training data paths provides a substantially higher return on computational investments compared to scaling parameter** bounds alone.

LLM GEMINI HAS BEEN USED TO CONVERT TEXT MATHEMATICAL FORMULA INTO LATEX CODE AND THE COMPARISON TABLE OF BASELINE FINDINGS REST OF THEM WERE WRITTEN MANUALLY